

HTML Notebook

FULL DETAIL EDITION

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HTML explained in *full* detail

Not just "here's a form" — every property, every input type, every variation, each with its own code and its own live output, so nothing is left out.

01 The Skeleton of a Page

Every HTML page is built on the same base structure. Here's what each part actually does:

TAG	WHAT IT MEANS
<code><!DOCTYPE html></code>	Tells the browser "this is a modern HTML5 page." Always the very first line.
<code><html></code>	The outer container for the entire page. Everything else lives inside it.
<code><head></code>	Holds information ABOUT the page — title, linked files, meta info. Nothing here is visible directly on the page.
<code><title></code>	The text shown on the browser tab.
<code><meta charset="UTF-8"></code>	Tells the browser which characters/letters the page uses so text renders correctly.
<code><body></code>	Everything visible on the page — text, images, buttons — goes here.

● CODE.HTML

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>My First Page</title>
  </head>
  <body>
    <h1>Hello, World!</h1>
  </body>
</html>
```

● OUTPUT

Hello, World!

Remember: `head` = invisible setup information, `body` = everything a visitor can actually see.

02

Comments

A comment is a note written inside the code that the browser completely ignores. It never shows up on the page. Use it to leave yourself reminders or explain tricky parts of your code.

● CODE.HTML

```
<!-- This is a comment, it will not show up -->
<p>Only this line is visible.</p>
<!-- TODO: add a footer here later -->
```

● OUTPUT

Only this line is visible.

03

Headings

HTML gives you six levels of heading, `<h1>` through `<h6>`. `<h1>` is the biggest and most important (usually the page's main title — use it only once per page), and each number after that is a smaller sub-heading, like chapter and sub-chapter titles in a book.

● CODE.HTML

```
<h1>Heading Level 1</h1>
<h2>Heading Level 2</h2>
<h3>Heading Level 3</h3>
<h4>Heading Level 4</h4>
<h5>Heading Level 5</h5>
<h6>Heading Level 6</h6>
```

● OUTPUT

Heading Level 1

Heading Level 2

Heading Level 3

Heading Level 4

Heading Level 5

Heading Level 6

Rule of thumb: use only `<h1>` per page, and don't skip levels just to make text smaller — pick the level based on structure, use CSS for size.

04

Text Formatting — every tag, one by one

HTML has many small tags for shaping text. Here is each one explained separately, not lumped together.

`<p>` paragraph

Wraps a block of normal text. Browsers automatically add space above and below it.

● CODE

```
<p>This is a paragraph of text.</p>
```

● OUTPUT

This is a paragraph of text.

`<br>` line break

Forces a new line inside the same paragraph, without starting a whole new block. It has no closing tag.

● CODE

```
<p>First line<br>Second line</p>
```

● OUTPUT

First line
Second line

`<hr>` horizontal rule

Draws a horizontal line, usually used to separate sections of content.

● CODE

```
<p>Above the line</p>
<hr>
<p>Below the line</p>
```

● OUTPUT

Above the line

Below the line

`<strong>` vs `<b>`

`<strong>` makes text bold AND tells the browser/screen-reader it's important. `<b>` only makes it visually bold, with no extra meaning.

● CODE

```
<strong>Important text</strong>  
<b>Just bold text</b>
```

● OUTPUT

Important text
Just bold text

 vs <i>

**** italicizes text and adds "emphasis" meaning (like stressing a word when speaking).

<i> just makes it visually italic — good for terms, names of books, etc.

● CODE

```
<em>Emphasized text</em>  
<i>Just italic text</i>
```

● OUTPUT

Emphasized text
Just italic text

<mark> highlight

Highlights text with a yellow background, like using a highlighter pen.

● CODE

```
<p>Please <mark>highlight this</mark> part.</p>
```

● OUTPUT

Please **highlight this** part.

<small> fine print

Makes text smaller — commonly used for disclaimers, fine print, or side notes.

● CODE

```
<p>Regular price. <small>Terms apply.</small></p>
```

● OUTPUT

Regular price. Terms apply.

 vs <ins>

**** shows a strikethrough for removed/deleted text. **<ins>** underlines newly inserted text. Handy for showing edits.

● CODE

```
<p>Price: <del>₹999</del> <ins>₹699</ins></p>
```

● OUTPUT

Price: ~~₹999~~ ₹699

<sub> vs <sup>

<sub> drops text slightly below the line (subscript), **<sup>** raises it slightly above (superscript). Useful for formulas or footnotes.

CODE	OUTPUT
<code><p>H<sub>2</sub>O and x<sup>2</sup></p></code>	H ₂ O and x ²

`<blockquote>` vs `<q>`

`<blockquote>` is for a longer quoted block (usually indented on its own line). `<q>` is for a short inline quote — the browser adds quotation marks automatically.

CODE	OUTPUT
<code><blockquote> Big things are expected of us. </blockquote> <p>She said <q>let's begin</q>.</p></code>	Big things are expected of us. She said “let's begin”.

`<pre>` and `<code>`

`<code>` marks inline computer code with a monospace font. `<pre>` preserves exact spacing and line breaks — great for showing blocks of code, like this whole notebook does.

CODE	OUTPUT
<code><p>Use the <code>alert()</code> function.</p> <pre> line one line two, indented </pre></code>	Use the <code>alert()</code> function. line one line two, indented

`<abbr>` abbreviation

Marks an abbreviation. The `title` attribute holds the full meaning, shown as a tooltip on hover.

CODE	OUTPUT
<code><abbr title="HyperText Markup Language">HTML</abbr></code>	HTML (hover it to see the full title)

All links use the same `<a>` tag, but the `href` value and a few extra attributes change what actually happens when it's clicked.

Basic link to another website

● CODE

```
<a href="https://example.com">
  Visit Example Site
</a>
```

● OUTPUT

[Visit Example Site](https://example.com)

`target="_blank"` – open in a new tab

Without this, the link opens in the same tab. Adding `target="_blank"` opens it in a brand-new tab instead.

● CODE

```
<a href="https://example.com" target="_blank">
  Open in a new tab
</a>
```

● OUTPUT

[Open in a new tab](https://example.com) ↗

Email link – `mailto:`

Opens the visitor's email app with an address already filled in.

● CODE

```
<a href="mailto:hello@example.com">
  Email us
</a>
```

● OUTPUT

[Email us](mailto:hello@example.com)

Phone link – `tel:`

On a phone, tapping this starts a call to the number.

● CODE

```
<a href="tel:+911234567890">
  Call us
</a>
```

● OUTPUT

[Call us](tel:+911234567890)

Jump link – link to a section on the same page

Using `#` plus an element's `id` scrolls the page down to that element, exactly like this notebook's sidebar does.

● CODE

● OUTPUT

```
<a href="#section2">Jump to Section 2</a>
```

...later in the page...

```
<h2 id="section2">Section 2</h2>
```

[Jump to Section 2](#)

download attribute

Instead of opening the file, the browser downloads it directly.

● CODE

```
<a href="report.pdf" download>
  Download the PDF
</a>
```

● OUTPUT

↓ [Download the PDF](#)

06

Images & Figures

 basic image

A self-closing tag (no separate closing tag needed).

ATTRIBUTE	WHAT IT DOES
<code>src</code>	Path or URL of the image file. Required.
<code>alt</code>	Text description shown if the image fails to load, and read aloud by screen readers.
<code>width</code> / <code>height</code>	Sets the display size in pixels.
<code>title</code>	Extra text shown as a tooltip on hover.
<code>loading="lazy"</code>	Tells the browser to only load the image when it's about to scroll into view — improves page speed.

● CODE

```

```

● OUTPUT

🐱 image renders
here

<figure> and <figcaption>

Groups an image together with its caption, so they're understood as one unit — like a picture with a label underneath in a book.

● CODE

```
<figure>
  
  <figcaption>Fig 1. A very sleepy cat.</figcaption>
</figure>
```

● OUTPUT



Fig 1. A very sleepy cat.

07

Lists — all three kinds

 unordered (bullet) list

Use when the order of items doesn't matter.

● CODE

```
<ul>
  <li>Tea</li>
  <li>Coffee</li>
  <li>Juice</li>
</ul>
```

● OUTPUT

- Tea
- Coffee
- Juice

 ordered (numbered) list

Use when order matters, like steps. **start** changes the first number, **reversed** counts backwards, and **type** switches between 1/A/a/I/i style numbering.

● CODE

```
<ol>
  <li>Preheat the oven</li>
  <li>Mix the batter</li>
  <li>Bake for 20 minutes</li>
</ol>

<ol type="A">
  <li>First option</li>
  <li>Second option</li>
</ol>
```

● OUTPUT

1. Preheat the oven
 2. Mix the batter
 3. Bake for 20 minutes
- A. First option
 - B. Second option

<dl> description list

Pairs a term (`<dt>`) with its description (`<dd>`) — great for glossaries or FAQs.

● CODE

```
<dl>
  <dt>HTML</dt>
  <dd>The markup language that builds web pages.</dd>
  <dt>CSS</dt>
  <dd>The language that styles web pages.</dd>
</dl>
```

● OUTPUT

HTML

The markup language that builds web pages.

CSS

The language that styles web pages.

08

Tables — full structure

Basic table: `<table>`, `<tr>`, `<th>`, `<td>`

`<tr>` = table row. `<th>` = header cell (bold, centered by default). `<td>` = normal data cell.

● CODE

```
<table>
  <tr>
    <th>Name</th>
    <th>Score</th>
  </tr>
  <tr>
    <td>Riya</td>
    <td>92</td>
  </tr>
</table>
```

● OUTPUT

Name	Score
Riya	92

`<thead>`, `<tbody>`, `<tfoot>` and `<caption>`

These group the header row, the main data rows, and a summary row separately — and `<caption>` gives the whole table a title.

● CODE

```
<table>
  <caption>Weekly Scores</caption>
  <thead>
    <tr><th>Name</th><th>Score</th></tr>
  </thead>
```

● OUTPUT

```
<tbody>
  <tr><td>Riya</td><td>92</td></tr>
  <tr><td>Aman</td><td>85</td></tr>
</tbody>
<tfoot>
  <tr><td>Average</td><td>88.5</td></tr>
</tfoot>
</table>
```

Weekly Scores

Name	Score
Riya	92
Aman	85
Average	88.5

colspan and rowspan

`colspan` makes a cell stretch across multiple columns. `rowspan` makes it stretch down multiple rows.

CODE

```
<table>
  <tr>
    <th colspan="2">Contact Info</th>
  </tr>
  <tr>
    <td>Phone</td>
    <td>123-456</td>
  </tr>
</table>
```

OUTPUT

Contact Info	
Phone	123-456

09 Forms — every property in full detail

A form isn't just one tag — it's a whole family of elements. Below is each one on its own: every common `<input>` type, plus `<select>`, `<option>`, `<textarea>`, `<label>`, `<fieldset>`, and every button type.

`<form>` the container

Everything a user fills in goes inside this tag. Its two key attributes:

ATTRIBUTE	WHAT IT DOES
<code>action</code>	The web address where the form's data gets sent when submitted.
<code>method</code>	<code>get</code> sends data visibly in the URL; <code>post</code> sends it hidden in the request body — used for anything sensitive, like passwords.

CODE

OUTPUT

```
<form action="/submit" method="post">
  ... all fields go here ...
</form>
```

An empty form has no visible shape by itself — its fields (below) are what actually render.

<label> labeling a field

Connects a text label to a field using matching `for` / `id` values. Clicking the label focuses the field — and it's essential for accessibility.

● CODE

```
<label for="uname">Username</label>
<input type="text" id="uname">
```

● OUTPUT

Username

Input types — the `type` attribute on `<input>` changes everything about how it behaves:

`type="text"`

A plain single-line text box. The most basic input type.

● CODE

```
<input type="text" placeholder="Your name">
```

● OUTPUT

Your name

`type="email"`

Same as text, but the browser checks it looks like a real email address (has an @) before letting the form submit.

● CODE

```
<input type="email" placeholder="you@example.com">
```

● OUTPUT

you@example.com

`type="password"`

Hides what's typed behind dots, for secret input.

● CODE

```
<input type="password" placeholder="Password">
```

● OUTPUT

.....

`type="number"`

Only accepts numbers, and shows small up/down arrows. `min` and `max` limit the range.

● CODE

● OUTPUT

```
<input type="number" min="1" max="10">
```

1

type="date", "time", "month"

Shows a native calendar / time / month picker — no extra code needed.

CODE

```
<input type="date">
<input type="time">
<input type="month">
```

OUTPUT

dd/mm/yyyy



--:-- --





type="checkbox"

A box that can be ticked on or off. Several checkboxes can be selected at the same time — good for "pick any that apply."

CODE

```
<input type="checkbox" id="c1">
<label for="c1">I agree to the terms</label>
```

OUTPUT



I agree to the terms

type="radio"

A round button where only ONE option can be picked. All radios that should be linked together share the same `name`.

CODE

```
<input type="radio" name="plan" id="basic">
<label for="basic">Basic</label>

<input type="radio" name="plan" id="pro">
<label for="pro">Pro</label>
```

OUTPUT

☒ Basic

☐ Pro

type="range"

A slider for picking a number between `min` and `max`.

CODE

```
<input type="range" min="0" max="100">
```

OUTPUT



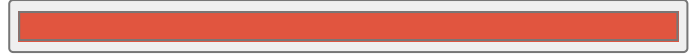
type="color"

Opens a native color picker and returns a hex color value.

● CODE

```
<input type="color" value="#E1553F">
```

● OUTPUT



type="file"

Lets the user pick a file from their device to upload.

● CODE

```
<input type="file">
```

● OUTPUT

Choose File No file chosen

type="search", "tel", "url"

Each looks like a text box but hints the right keyboard on mobile and adds light validation — `tel` for phone numbers, `url` checks for a valid web address pattern.

● CODE

```
<input type="search" placeholder="Search...">
<input type="tel" placeholder="Phone number">
<input type="url" placeholder="https:// ...">
```

● OUTPUT

Search...

Phone number

https://...

type="hidden"

Completely invisible to the user, but still sent with the form — used to pass along data like an ID the page already knows.

● CODE

```
<input type="hidden" name="userId" value="4521">
```

● OUTPUT

(nothing shows on screen — it's hidden by design)

<textarea> multi-line text

Unlike `<input>`, this is its own tag (not a type of input) because it needs an opening and closing tag to hold multiple lines. `rows` and `cols` set its visible size.

● CODE

```
<textarea rows="3" placeholder="Write your message..."></textarea>
```

● OUTPUT

Write your message...



<select> and <option>

A dropdown menu. `<select>` is the outer box, and each `<option>` is one choice inside it.

Add `selected` to an option to make it the default.

● CODE	● OUTPUT
<pre><select> <option value="in">India</option> <option value="us" selected>USA</option> <option value="uk">UK</option> </select></pre>	

`<optgroup>` grouping dropdown options

Groups related options under a bold sub-heading inside the dropdown.

● CODE	● OUTPUT
<pre><select> <optgroup label="Fruits"> <option>Apple</option> <option>Mango</option> </optgroup> <optgroup label="Vegetables"> <option>Carrot</option> </optgroup> </select></pre>	

`<fieldset>` and `<legend>`

Draws a box around a related group of fields, with `<legend>` as its title — useful for breaking a long form into clear sections.

● CODE	● OUTPUT
<pre><fieldset> <legend>Contact Details</legend> <input type="text" placeholder="Name"> <input type="email" placeholder="Email"> </fieldset></pre>	

Buttons — three different jobs, three different `type` values:

`type="submit"`

Sends the whole form's data to the server.

● CODE	● OUTPUT
<pre><button type="submit">Submit</button></pre>	

type="reset"

Clears every field back to its original value.

CODE

<button type="reset">Reset</button>

OUTPUT

Reset

type="button"

Does nothing on its own — it's a plain clickable button meant to be wired up with your own JavaScript.

CODE

<button type="button">Click Me</button>

OUTPUT

Click Me

Field-level attributes that apply to most inputs

ATTRIBUTE	WHAT IT DOES
placeholder	Faint hint text shown inside an empty field; disappears once typing starts.
required	Stops the form from submitting until this field is filled in.
disabled	Greys the field out — it can't be clicked, typed in, or submitted.
readonly	Can be seen and selected, but not edited.
value	Sets the field's starting value.
name	The key used to identify this field's data when the form is submitted.

CODE

<input type="text" value="Locked value" readonly>
<input type="text" placeholder="Can't touch this" disabled>
<input type="text" required placeholder="Required field">

OUTPUT

Locked value

Can't touch this

Required field

`<div>` is a block-level box for grouping bigger chunks of content — it always starts on a new line. `<span>` does the same job but inline, for wrapping a small bit of text without breaking the line. Neither has any look of its own — they exist purely so CSS or JavaScript can target that group.

● CODE.HTML

```
<div style="background:#eee; padding:10px;">
  This whole box is a div.
  <span style="color:red">
    Only this bit is a span.
  </span>
</div>
```

● OUTPUT

This whole box is a div. **Only this bit is a span.**

11

Semantic Layout Tags

These tags describe what a section of the page actually MEANS, instead of using a generic `<div>` for everything. This helps browsers, screen readers, and other developers understand the page's structure at a glance.

TAG	MEANING
<code><header></code>	Introductory content — usually a logo, title, or top navigation.
<code><nav></code>	A block of navigation links.
<code><main></code>	The primary, unique content of the page (only one per page).
<code><section></code>	A distinct grouped section of content, usually with its own heading.
<code><article></code>	Self-contained content that could stand alone, like a blog post or news story.
<code><aside></code>	Content related to but separate from the main content, like a sidebar or a tip box.
<code><footer></code>	Closing content — usually copyright, links, or contact info.
<code><details></code> / <code><summary></code>	Creates a native expand/collapse widget — no JavaScript required.

Header, nav, and footer together

● CODE

● OUTPUT

```
<header>Site Logo</header>
<nav>Home | About | Contact</nav>
<main>Page content goes here</main>
<footer>© 2026 My Site</footer>
```

Site Logo

Home | About | Contact

Page content goes here

© 2026 My Site

`<details>` / `<summary>` – expand and collapse

Try clicking it below — it actually works, with zero JavaScript.

CODE

```
<details>
  <summary>Click to expand</summary>
  <p>Here is the hidden content!</p>
</details>
```

OUTPUT

▶ Click to expand

12

Global Attributes

These attributes can be added to almost any HTML tag, no matter what it is.

ATTRIBUTE	WHAT IT DOES
id	A unique name for one specific element — used for jump links and targeted styling. No two elements should share an id.
class	A label that can be reused on many elements — the main way CSS targets groups of elements for styling.
style	Applies CSS directly to just that one element, right inline.
title	Extra info shown as a tooltip when hovering.
data-*	A custom attribute for storing your own extra information on an element, for JavaScript to read later — e.g. <code>data-user-id="42"</code> .

CODE.HTML

```
<p
  id="intro"
  class="highlight"
  title="A friendly hello"
  data-user-id="42"
```

OUTPUT

Hover me to see the tooltip.

```
style="color:#E1553F;"
>
  Hover me to see the tooltip.
</p>
```

id vs	think	id	as a person's name (one specific	class	as a job title (many people can
class:	of		person) and		share it).

Made as a full-detail study notebook — open this file in any browser and scroll through at your own pace. Every code block on this page reflects exactly what's rendered next to it.